

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 9

Topology: Structure Without Measurement

Exercises

Self-Study Curriculum

1. Consider the letter of the alphabet A. How many “holes” does it have topologically? What about B? D? O? Which letters are topologically equivalent to each other?

2. The “parameter space” of a logistic regression model with two features is $\mathbb{R}^2$. The parameter space of a logistic regression with two features and a constraint that the parameters sum to 1 is a line in $\mathbb{R}^2$. What is the topology of this constrained parameter space? Is it compact? Connected? What are the practical implications (e.g., for optimization)?

3. You are studying a dynamical system and you want to know whether it has a stable fixed point. The Brouwer Fixed Point Theorem says: every continuous function from a compact convex set to itself has a fixed point. This is a purely topological result. What are its assumptions, and why does each one matter? Can you think of a system where each assumption individually fails?

4. A colleague proposes that two datasets have “the same shape.” What does this mean topologically? How would you test it using TDA? What would it mean for two datasets to have different

shapes?
